🍄 Peter McCoy – Fungi in Soil Ecology

Sustainable Design Master Class | Host: Riley Latham
Video: YouTube | ~60 min
Book: Radical Mycology (Peter McCoy)

1. Title Slide

⏱ **00:00 – 03:00**

> So hey everybody I'm Riley Latham, welcome back to those who are joining us for the sustainable design master class. We're all about bringing some of the top regenerative designers in the world along with scientists, farmers and entrepreneurs to give you a picture of some of the best pathways for the future — whether that's improving your land, your business or your life.
>
> So today Neil Spackman alas he can't be here till an hour into the QA, he's meeting with some hotshot Saudi investors for Al Bayda which is a good thing. But I'll be taking his place. Just reminding you guys — turn off your Facebook, I know you guys are on Facebook, turn off

your phones because there's gonna be a lot of information coming away and if you want to get the most out of it you want to be present for this.

So today we're joined by Peter McCoy of Radical Mycology. Peter has made it his life's work to study almost a lost art of fungi, of mycology, because below all our feet we are surrounded by the kingdom of fungi. It is a thing that unites all soils, all trees, all ecology and yet we know so little about it. It has the ability to impact every aspect of our lives — it can make the difference between failure and survival in the driest parts of the world or the wettest parts of the world.

So I'm stoked to bring on Peter McCoy because he's going to talk about how mycology can be used as a tool in our soils and in our ecology. So without further ado I'm going to turn it over to Peter. Welcome to the webinar Peter.

Yeah thanks Riley for having me, thanks everybody around the world and into the future watching this. I'm incredibly excited to talk about this topic — I don't think I've ever actually done a talk on it specifically and there's so much to say about this huge black box of the soil. But even the fungal component which is a significant aspect of it and mycology in general as Riley said is so commonly overlooked and just poorly understood generally, especially amongst — well not necessarily especially but including — soil biologists, permaculture, those people in regenerative agriculture. It's just this big missing piece of the puzzle that so much of my work revolves around exploring, advocating for and raising awareness around.

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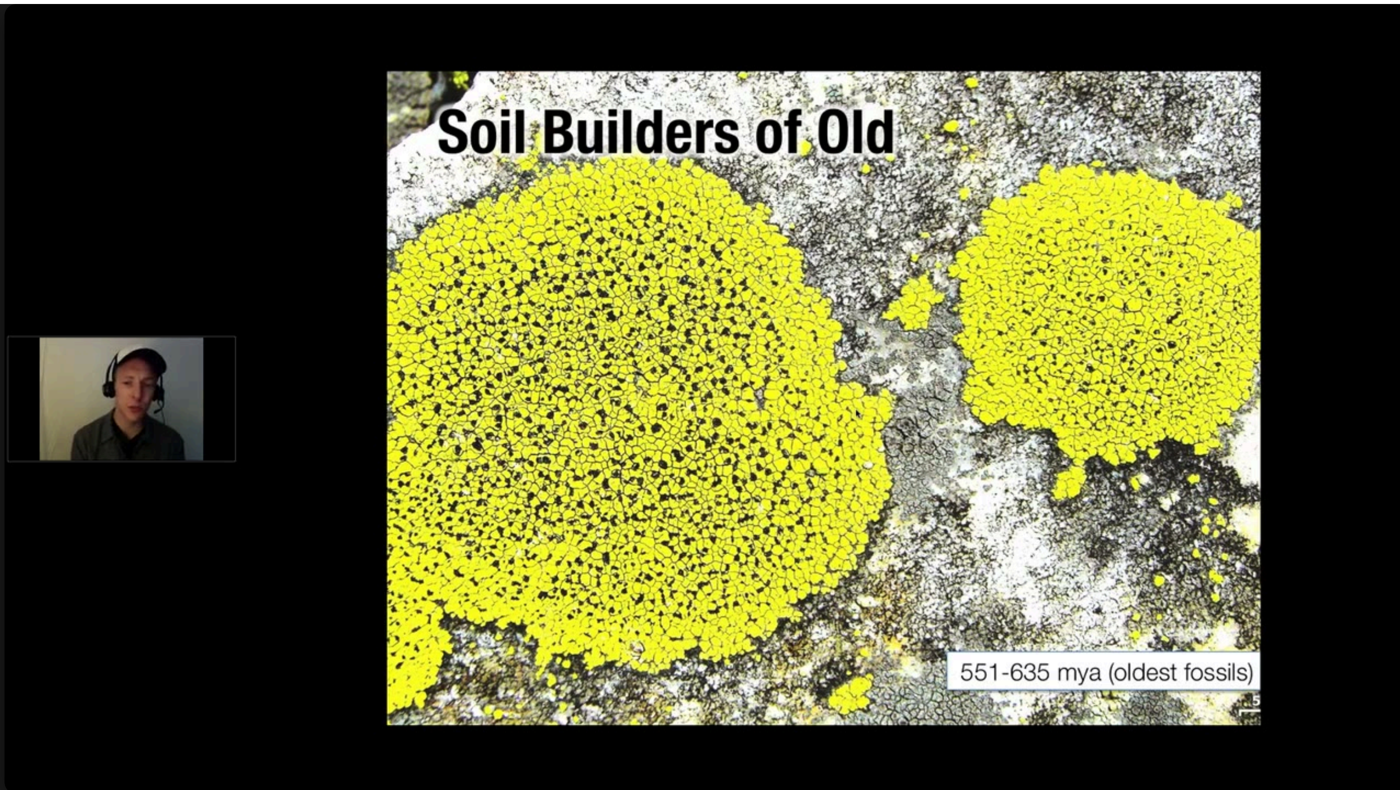
2. Soil Builders of Old



⏱ 03:13 – 07:04

One thing I think it's really important to point out up front is that mycology, the study of fungi, is actually considered a neglected mega science amongst academics. On one hand we nowadays know after really only about a hundred years of good study — even 50 years really looking at fungal ecology — that fungi are integral to pretty much every aspect of the natural world directly in many ways, sometimes indirectly. And yet we know so little and also so few people are actually studying it actively academically.

And then at the same time on the street level you get very few books enabling people to study the topic. So by and large it's very inaccessible — most people don't know much about it, we don't learn much about it in our schooling. So I always put that out up front to point out that generally just culturally we have this awareness gap and it's nobody's fault but something we need to be aware of and work collaboratively to overcome.

Because we're missing out on so much. Over the years of human civilization expansion we've probably impacted fungal ecologies in ways we've really yet to measure and understand. We measure flora and fauna but we never really look at the fungi. Even today in environmental surveys most agencies don't consider the fungal component of an ecosystem.

It's really important to recognize that essentially fungi have been the soil builders of the planet essentially since the beginning of the earth. Long story short — evolutionary biologists have argued that the first larger complex organisms on the early Earth on the supercontinent which was basically sterile rock — when it first formed, after bacteria formed a couple of millions of years later lichens formed.

And lichens today are the first things that show up after a lava flow or landslides. The fungal component of lichens — lichens are 95% fungal tissue — that fungal component releases really strong acids that can release minerals from the rock, solubilize them. And also as they decompose, lichens themselves decompose over time, they become sort of organic matter. And really the first soils of the whole earth were built by lichens and primarily the fungal component of that.

So they've built the soils of the world since prehistory. They enabled plants to survive, develop roots. They built the soils back then and they still do so much today in this regard.

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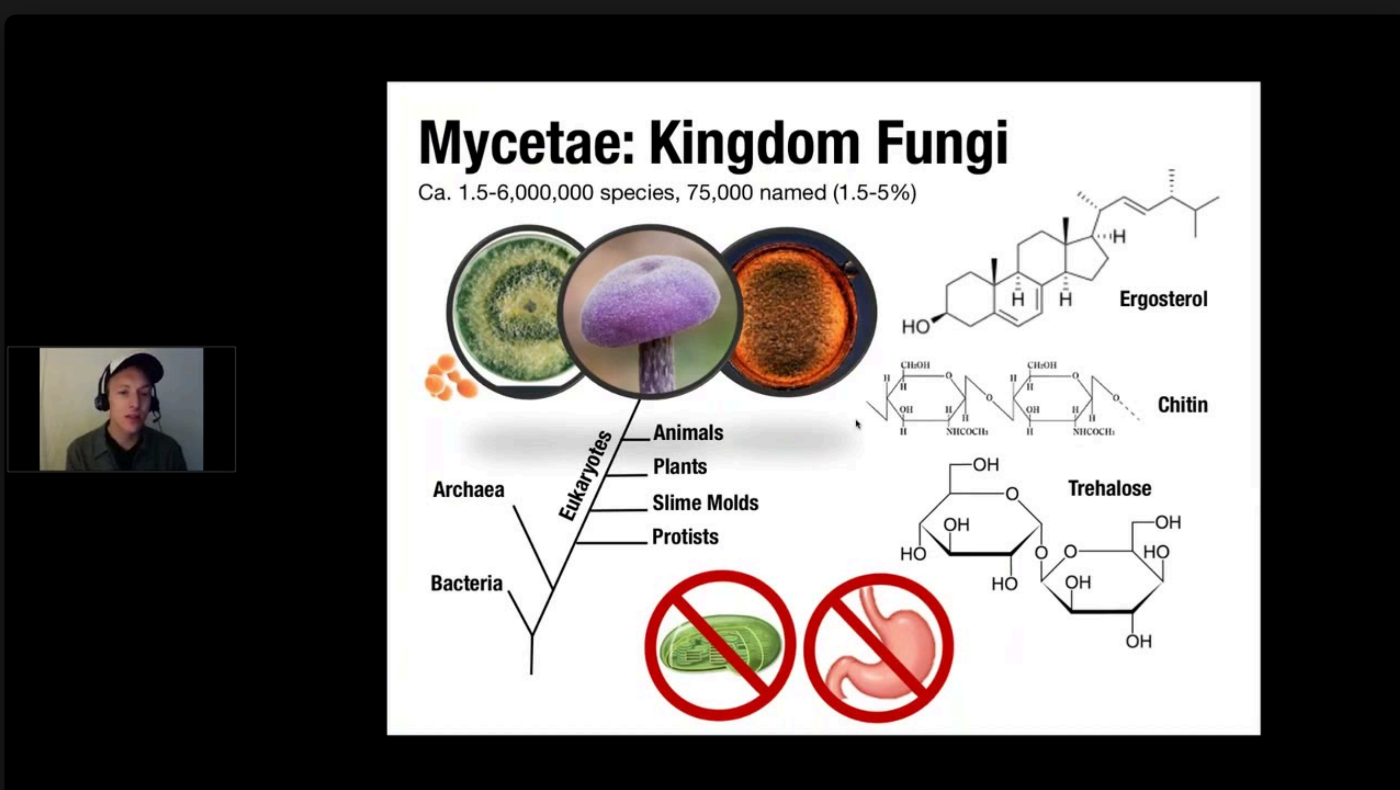
3. Mycetae: Kingdom Fungi



⏱ 07:10 – 11:00

So what are fungi? Fungi, the fungal kingdom or the Mycota, is a group of eukaryotic organisms — so they have nuclei in their cells, they're separate from bacteria. And they are distinguished from animals and plants in several ways. On one hand they have a unique component in their cell wall called ergosterol — it's sort of similar to our cholesterol but it's just unique to fungi. Their cell wall is not made out of cellulose like it is in plants but primarily made out of chitin — about 10 to 20 percent chitin and 80 to 90 percent high weight sugars, some of those sugars can be medicinal.

They store their energy as a form of sugar known as trehalose. But more importantly, more interesting for us, is the way that they consume. They do not have a stomach and they do not photosynthesize. Fungi sort of like bacteria get their nutrition through external digestion — they release all kinds of acids and enzymes that are uniquely tailored to their environment. They constantly respond to their environment and based on what is in their way essentially they will adjust the digestive enzymes that they need to eat those things — sort of simply put, break down that food externally and then consume the byproducts of it through their tissue through absorption.

Within the fungal Kingdom there's actually several subcategories — just like there's different types of animals there's many different types of fungi but we're going to focus on three types. On the left we have a mold, a Trichoderma mold, as a representative of a group known as the Ascomycota — this is our largest group of fungi. It includes many molds but also some mushrooms and yeast as well. Yeast only make up about 1% of the whole fungal Kingdom so they're a very small portion — the vast majority form networks of tissue.

On the right we see a large spore and that's representing a group known as the Glomeromycota — these are soil-dwelling fungi that don't form mushrooms but they're very very important for soil ecology. They're arguably the most ecologically important of all fungi. And then in the middle of course our beautiful mushrooms — the Basidiomycota — this is the category that contains the vast majority of our more commonly cultivated, wild harvested, edible and medicinal mushrooms.

Also to note is that the fungal Kingdom is incredibly large. Whereas there's about 300,000 plants in the world and we've named about 95% of them, estimates of the fungal Kingdom go anywhere from 1.5 to 6 million species and we've named only about 75,000 or 1.5 to 5 percent. So in essence we know so little about fungi, we've named so few of them — and that's just naming them, let alone understanding their ecological function.

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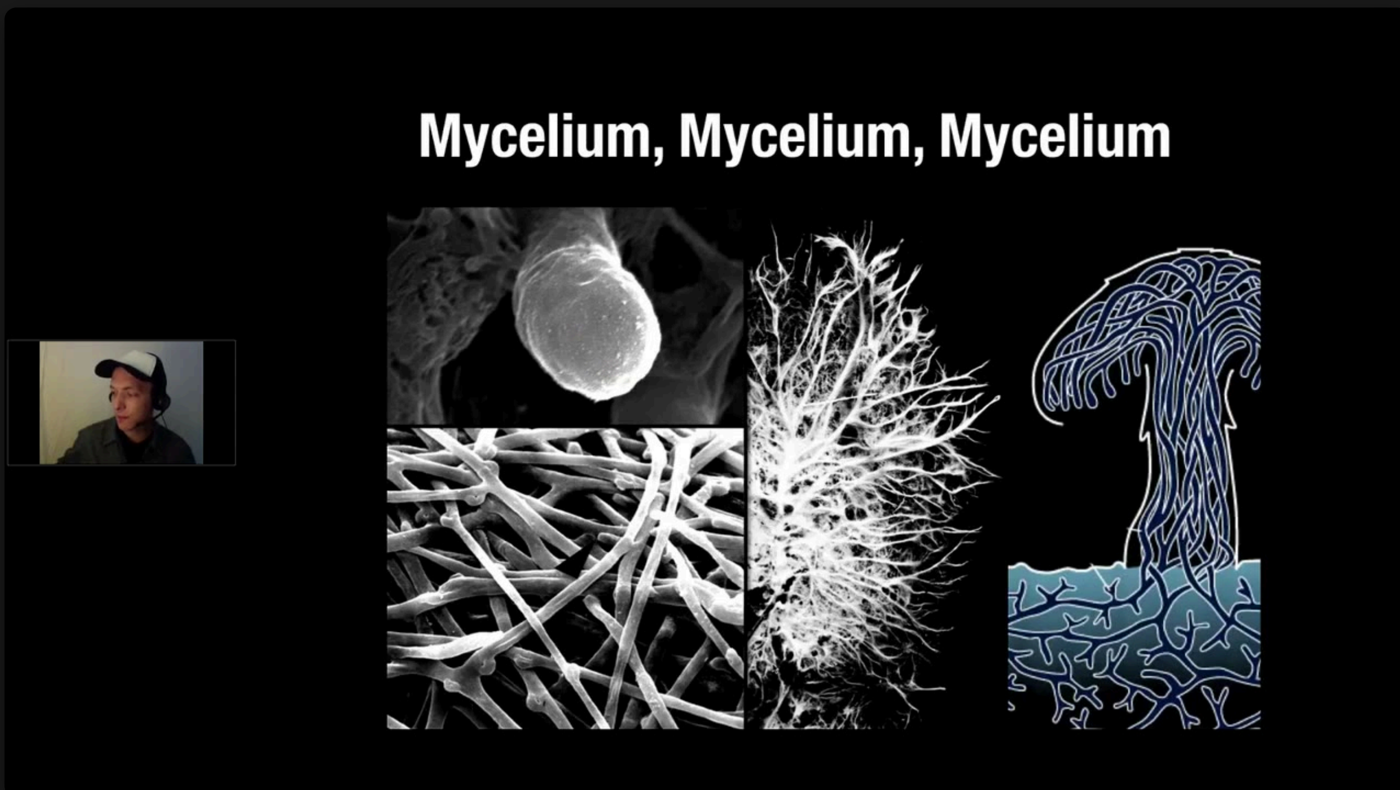
4. Mycelium, Mycelium, Mycelium



⏱ **11:00 – 12:54**

Mycology being a young neglected science — it's only about two hundred years old and we've barely scratched the surface really, largely in this ecological perspective.

Of the fungal Kingdom only about 1% form as yeast. And yeast are important in the soils — they likely contribute to nutrient cycling and they're actually probably a good food source for a lot of microbes and insects. But soil yeasts are pretty poorly understood.

The vast majority of fungi we're mostly going to focus on are those that form networks of tissue — that's known as mycelium. Mycelium is interesting biologically — in essence it's a network of one cell thick threads, filaments of tissue that grow from their tip, constantly responding and engaging with the environment at that tip. That's where all the digestive enzymes and acids are released and the byproducts of the digestion are consumed. And that's really what the fungus is — most fungi don't form mushrooms. They live as a network of mycelium like a mold, fanning through the soil, fanning through the wood doing all this interesting work.

This is really what the fungus is. Some of them produce different-looking spores, we call them different species, but fundamentally this is the primary fungal tissue. Of course some fungi condense that mycelium into a macro structure we call a mushroom — but even that is just condensed mycelium, it's not different tissue, not a different organ. It's all mycelium and that's where all the business happens.

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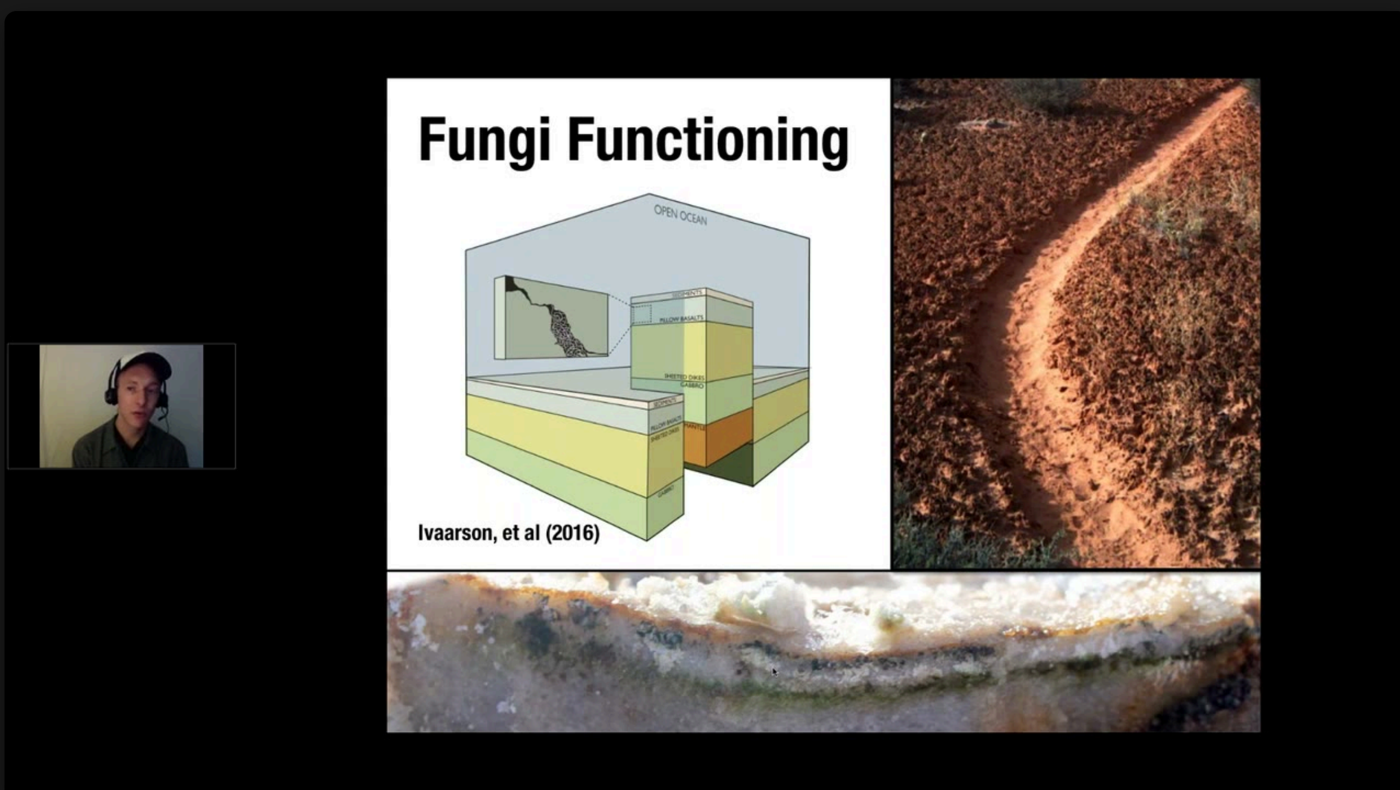
5. Fungi Functioning



⏱ 12:54 – 16:03

Where do we find fungi? Fungi live in all environments. On the upper left is a photo from a study that came out earlier this year essentially arguing that of what we know of marine fungi it seems highly plausible that mycelium permeates the entire ocean floor — which is the world's largest habitat. And just as on land, it's very likely that the mycelium is what takes in the dead ocean life and turns it up and breaks it down and recycles those nutrients. And of course a lot of those nutrients end up back on land. So fungi are fundamentally at the bottom of much of the world's food web — they do this nutrient cycling role as one of the primary functions.

On the upper right we see cryptobiotic crust community in desert ecosystems. They're very important in these dryland environments because they stabilize soils. These are communities of fungi, lichens, cyanobacteria, micro plants — and they basically create micro pits and valleys that can collect water, hold habitats and space and shelter for seeds, enabling them to establish in these environments. And they actually fix a lot of nitrogen for these ecosystems and the fungal component helps translate and translocate that nitrogen throughout those ecosystems.

These really complex communities take a very long time to develop — that's why you often see signs saying "don't bust the crust" because they take hundreds of years to develop. Much of our desert ecosystems in North America have had these communities destroyed by human intervention and animal compaction.

At the bottom is a photo of the ground crust of Antarctica — just below the surface layer is a black layer of melanized fungi that filter the intense UV of the sunlight in Antarctica for the plant component, the algal layer right below it. These are known as crypto endoliths. There are about 2 flowering plant species and over 430 lichens in Antarctica — much of them look like this, hiding in the crust layer.

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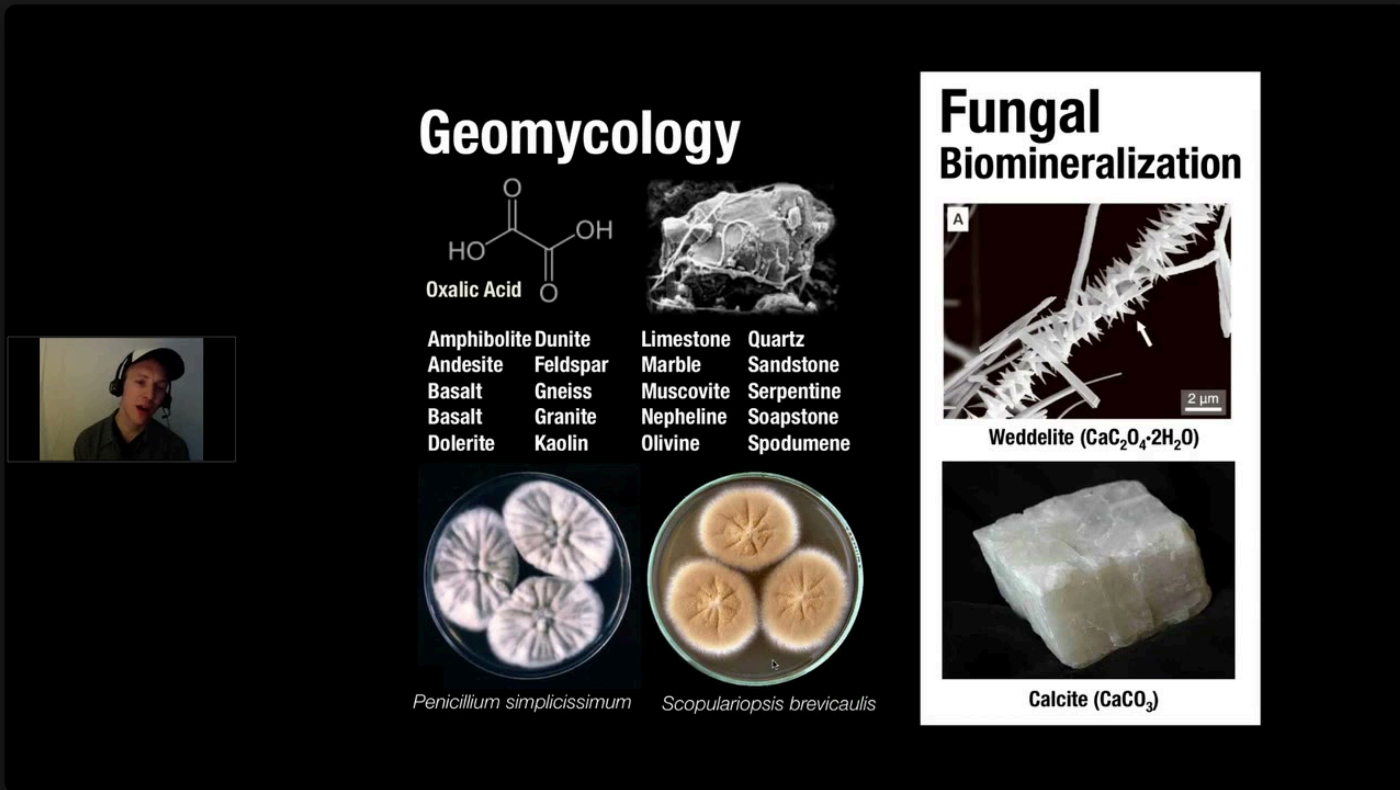
6. Soil Layers

⏱ **16:03 – 17:14**

> We're more focused today on the terrestrial environment and our terrestrial soils. Soils form sort of from the top and from the bottom. Down below there is the parent material, bedrock, that is slowly over the eons worn down into smaller and smaller mineral bits. Part of that we think is through of course weathering and some microbes can slowly break down minerals, but so do fungi — they very much are involved in releasing minerals from clay and rock and I think play a big part in this slow dissolve of parent material into the mineral component of soils from below.
>
> And then up top we have the organic material that falls down from plants and everything else and that slowly degrades into the more organic component. Fungi are integral to that decomposition. And in the middle we get a gradiation of the blend of the two — different types of soils are basically different ratios of these components.
>
> So how do fungi contribute to both ends of the spectrum — the mineral end and the organic end — and what do they do once that soil is formed? This is what we're focusing on today.

7. Geomycology & Fungal Biomineralization



⏱ 17:14 – 20:48

On the mineral end, fungi on one hand both dissolve rock — break it down, release those minerals — and on the other end they can actually recombine certain elements and create minerals in the soil through biomineralization.

On the left we're looking at a topic known as geomycology — a term coined by a researcher by the name of Gadd. Many types of fungi, many micro fungi and molds, have been shown to break down all kinds of rocks. In the middle we see many types of rock that fungi have been shown to grow on and release minerals from — primarily through the release of really strong acids from the tips of that mycelial network. Oxalic acid is a common one, malic acid, citric acid and a few others — really strong acids that can release these minerals quite well into the environment both for the fungus but also for the plants it associates with.

The two molds on the bottom — Penicillium simplicissimum and Scopulariopsis brevicaulis — have been shown to release aluminum from aluminosilicate clays. So just like some bacteria have been known to solubilize and release minerals, fungi do it as well but as with so many areas of mycology this is just really poorly looked at.

And at the same time they help recombine minerals or elements in the soil and create different sinks. Some strong examples are the combination of calcium and carbon into different compounds like weddellite and whewellite — these are two really common carbon minerals in the soil, some of the biggest carbon sinks. These are likely produced by fungal biomineralization. In the top picture we're actually seeing a fungal hyphal thread covered in weddellite crystals — pretty amazing photo. And a new paper just came out talking about how it's very likely that fungi produce calcite — another important calcium and carbon sink in the soil.

As my friend says, fungi are nature's greatest chemists — they do so many things chemically that are unparalleled by bacteria, animals and plants.

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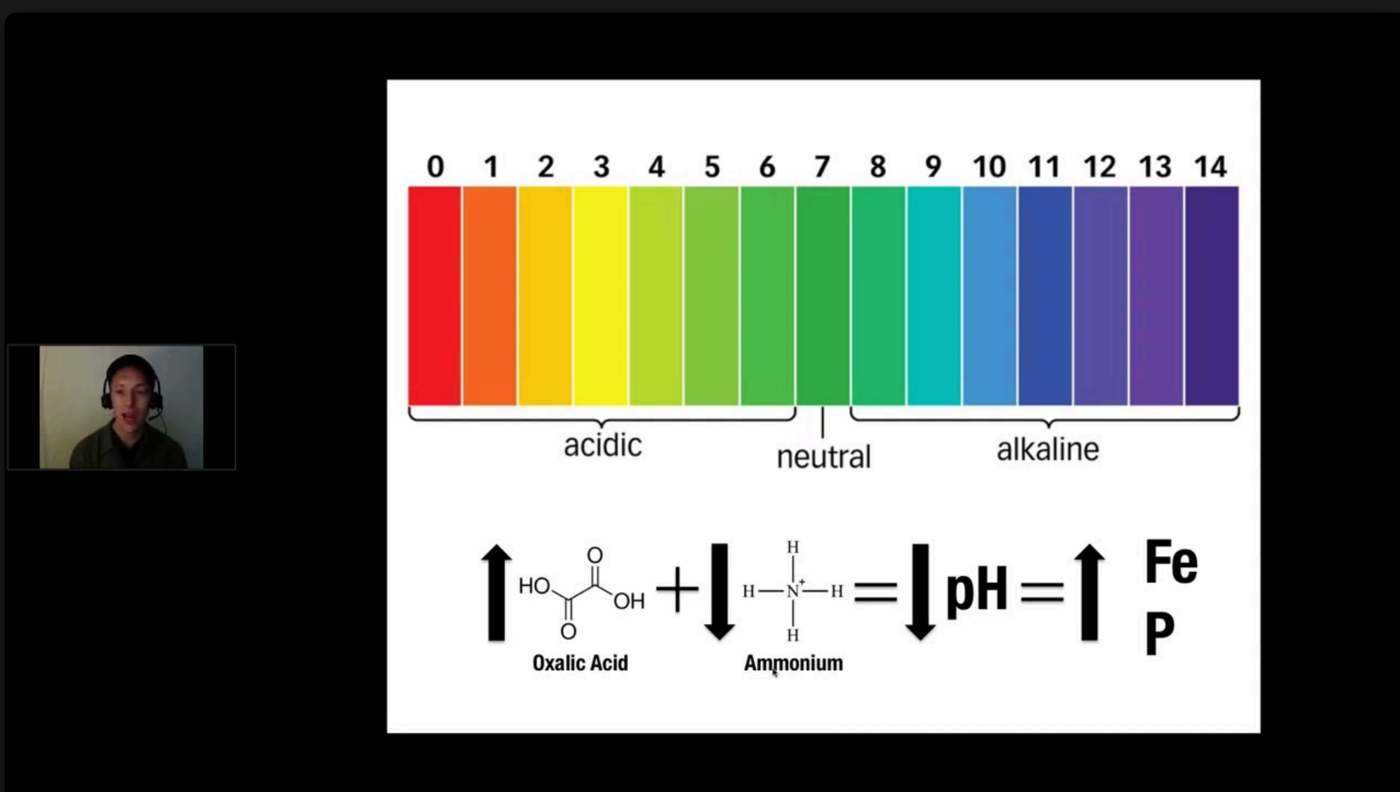
8. pH Scale – Fungi Influence Soil pH



⏱ 20:48 – 21:56

Fungi also influence soil pH in a variety of ways. One common example is that various types of soil fungi release oxalic acid which of course is acidic — so the arrow is going up — and they can also uptake ammonium which also contributes to the lowering of pH. And in lower pH environments many minerals and metals and trace elements are more available to plants.

So fungi help lower the pH — this is why fungal soils are a bit more acidic. They contribute to that acidification so that plants can get the nutrients they need. Arguably if you want to think about it in this sort of meta picture — fungi sort of sculpt our environments, they really contribute to the flow of nutrients and they fundamentally determine what plants and ultimately animals can live in an environment.

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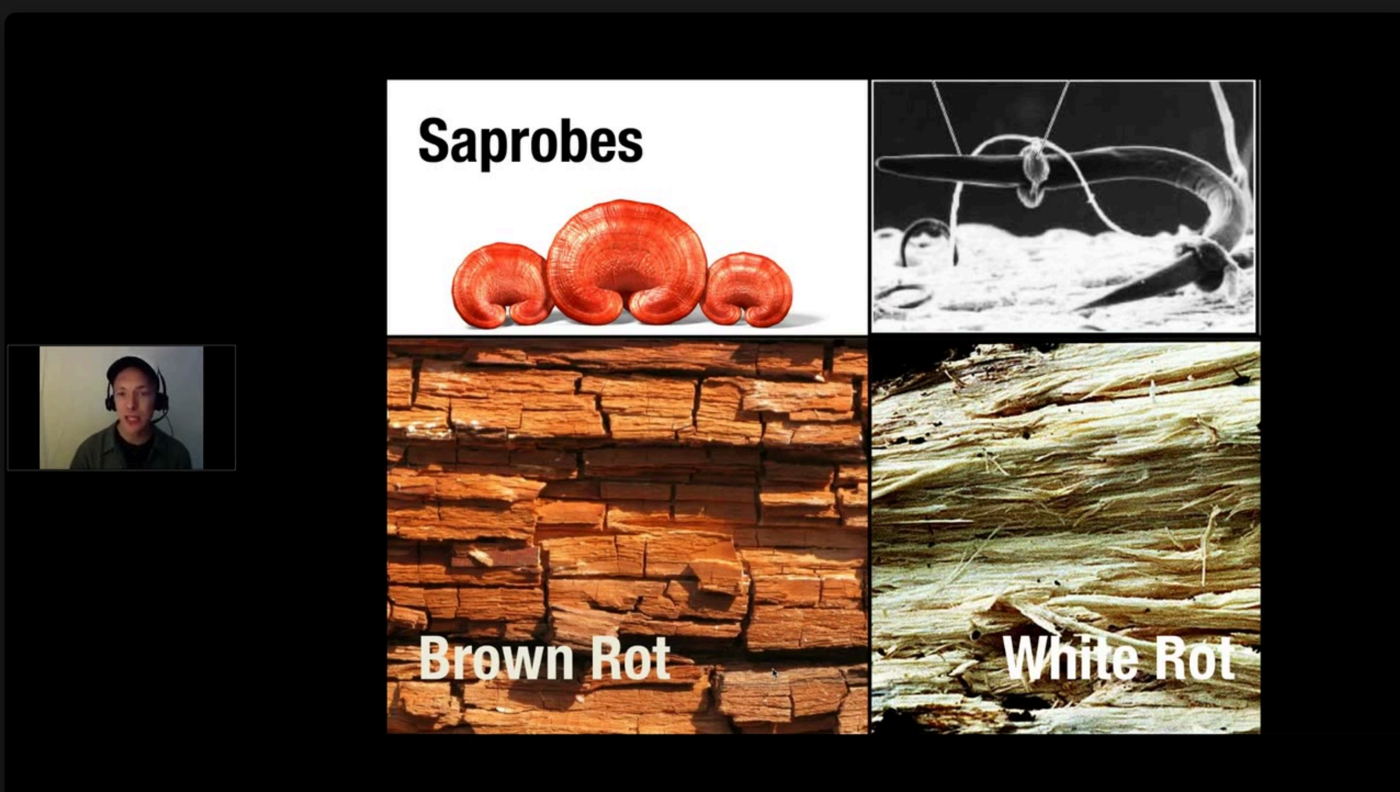
9. Saprobes – Decomposers



⏱ **21:56 – 24:45**

On the other end we have the decomposing fungi that break down the organic matter. Fungi are responsible for 90% of decomposition on the planet and bacteria primarily fill in the gaps — they really eat the smaller stuff that fungi have released from the more complex compounds. Bacteria can readily consume things like sugars when they're available but stuff that's locked up in lignin — which is nature's most complex compound — only fungi can break down lignin. Lignin is what makes wood hard and rigid and the fungi make that available.

On the upper left — a beautiful reishi mushroom. Many of the mushrooms we cultivate commercially for food and medicine are decomposers, they don't have symbiotic relationships, they just eat dead things. Some fungi can even attack living organisms like nematodes — some will actually eat micro insects like springtails and suck out their life and feed that through to their tree partners.

Wood decomposers are a major piece of the soil building puzzle. There's two main types. On the left the brown rotters that break down cellulose and hemicellulose, leaving behind the blocky lignin. This brown rot material — cubical brown rot, CBR — might actually be a really important component of soils. It's like a sponge, kind of nature's biochar as my friend says — holds a lot of water, nutrients, roots will grow into it, microbes inhabit it.

On the right we have the white rot which is produced by the only fungi that can break down lignin — about 2,000 species of fungi. They eat the lignin, leave behind the stringy cellulose. And this has actually been argued to perhaps contribute to the production of some of the humic acids and humic substances in our soils.

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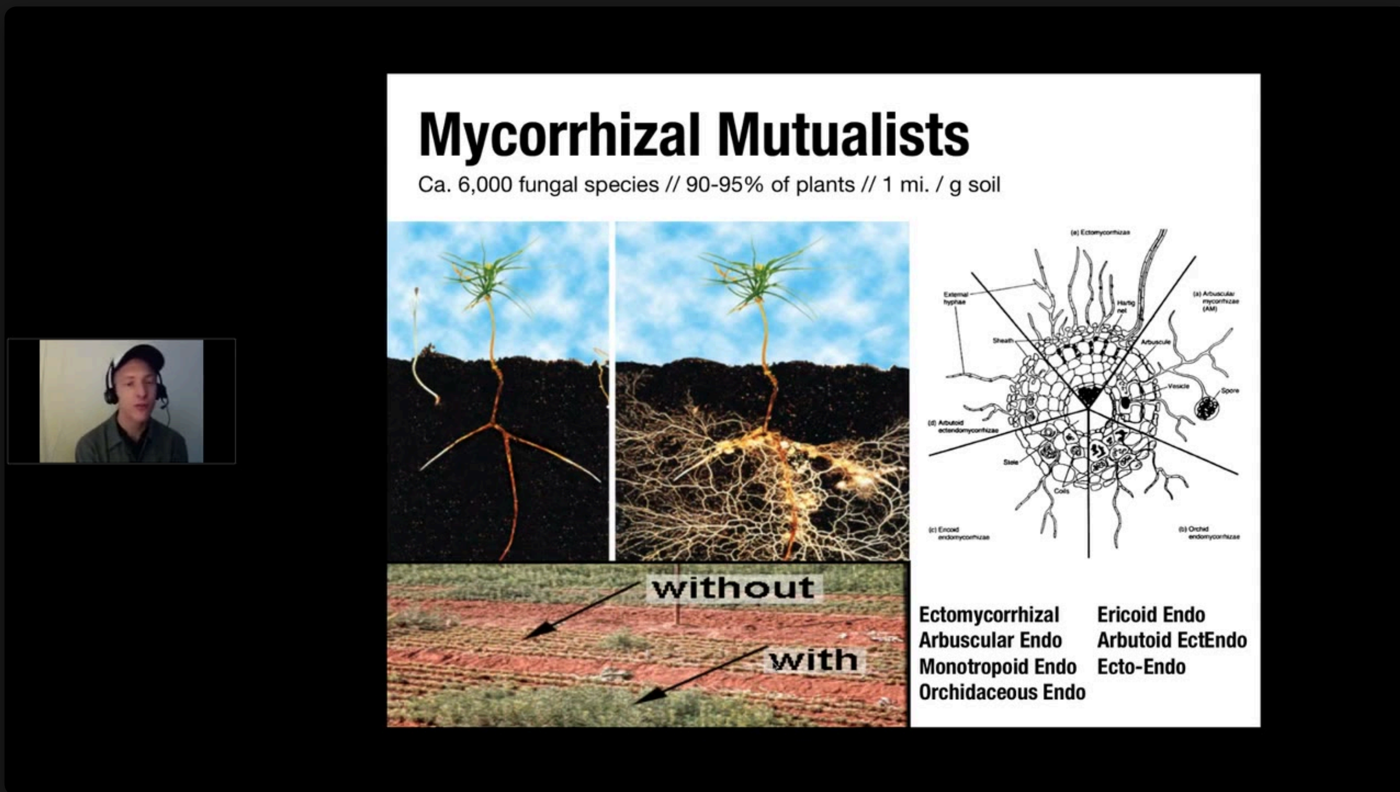
10. Trichoderma spp.

⏱ **24:45 – 25:59**

> One to be aware of for gardeners are the Trichoderma species — these are a very big genus of fungi around the world. Several species have been heavily investigated for their ability to primarily defend against other fungi. These are actually a major check-and-balance in the soil — even though they don't look so pretty, they're actually very important for fighting off other fungal pathogens.
>
> You can easily grow this mold — it grows on almost anything sugary — and then you can apply it to the root zones of plants and they will defend against pathogens. Of course you have to try to bring that in before the pathogen gets there — if the pathogen's already taken over it's kind of hard for the Trichoderma to win, but it's always worth the shot if you're in a kind of do-or-die scenario.
>
> They can also be added to compost piles and actually speed up the decomposition quite rapidly — they have really strong cellulases, they break down plant matter very quickly. There's actually a lot of research been done with Trichoderma in both these regards.

11. Mycorrhizal Mutualists



⏱ 25:59 – 29:00

We're mostly going to focus on the fungi that are perhaps the most studied in soil systems — the mycorrhizal species. Mycorrhiza meaning fungus, rhiza meaning root. These are fungi that symbiotically form a relationship with plant roots, clamping onto the roots and exponentially increasing their surface area 10 to a thousandfold.

The onion-inoculated plant with its roots can't really access a lot of nutrients. Roots in many ways are like porous anchors — they hold the plant in place but they haven't evolved to do the chemical work needed to transform nutrients and bring them in. They've primarily evolved to rely on fungi and other microbes to do this work for them.

The oldest plant fossil with roots has this relationship from about 450 million years ago. And today 90 to 95% of all plants form or can form this relationship. Many of them — like melons, oak, pines, citrus plants, grapes — will not survive in the wild without these mycorrhizal partners. In our crop lands we have to heavily fertilize and provide all kinds of defense chemicals to sort of try to mimic what the fungus naturally does.

Many of our crops will significantly benefit from the introduction of these fungi because they not only fan out and access nutrients and do all this chemical work that the plant just can't do

— they also draw on water enabling greater drought resistance. And even beyond that, they connect many plants together. In an intact forest system one mycelial network could connect potentially hundreds or thousands of species of plants together in what we call a common mycelial network, a CMN — and intelligently distribute nutrients amongst those organisms in ways that keeps everything going and everything alive.

This is how the baby trees survive in the depths of the forest — it's the mycorrhizal fungi providing sugars from the mother tree that's reached the canopy. There's other studies that have shown that when one plant gets hit with aphids, the mycorrhizal fungi will translate some sort of information to other plants in the system and those plants will start to produce anti-aphid compounds.

There's only about 6,000 fungal species that form this relationship that we know of. In a healthy soil system every gram of soil might have a mile or so of fungal mycelium — very efficiently packed, very high surface to volume ratio.

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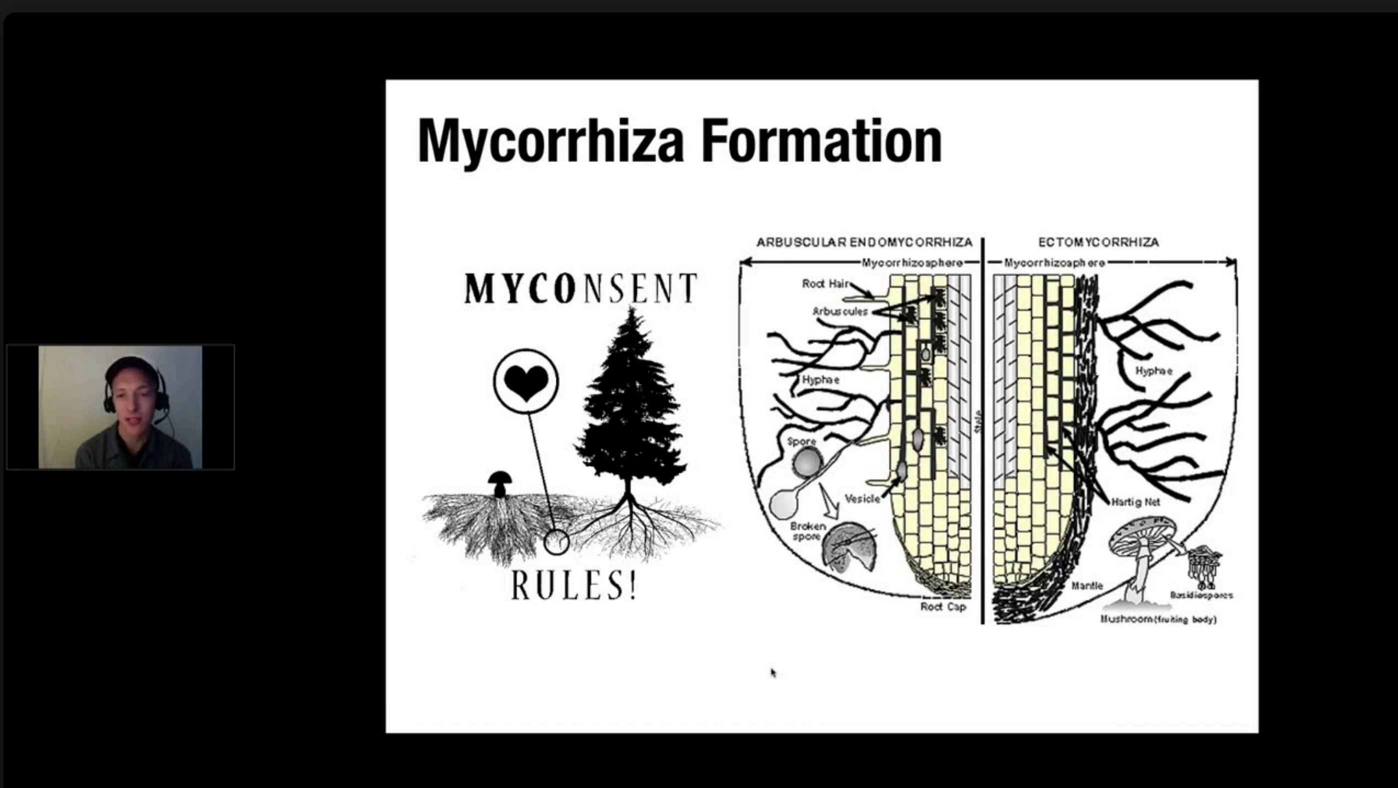
12. Mycorrhiza Formation



⏱ **29:00 – 31:06**

> There's about seven types of mycorrhizal fungi that we differentiate — based on the structures they form, how deep they go into the root, and if they penetrate into the root cell wall itself.
>
> The formation seems primarily to happen when the plant is starved for nutrition. If the plant is overly fertilized it doesn't need the fungus — so if you're dumping inputs into the soils then the relationship won't form. If you buy products that have mycorrhizal fungi but they're also nutrient rich, it's kind of a marketing scheme that isn't necessarily going to benefit you. Primarily it seems to happen when the plant is starved especially of phosphorus and nitrogen.
>
> Under those stressful conditions the spores need to be introduced to the root zone of the plant — you can't top dress the soil — the spore will germinate, there'll be a communication, and the plant allows the fungus in.
>
> Some fungi only form on the surface of the root and form a thicker layer — only go a couple layers deep into the root — these are known as ectomycorrhizal fungi on the far right. They don't necessarily penetrate into the root cell itself.
>
> On the left we have the representative of the endomycorrhizae fungi — these usually go deeper into the root but more specifically actually penetrate through the cell wall (not the cell membrane) and form different structures to exchange nutrients with the plant partner.

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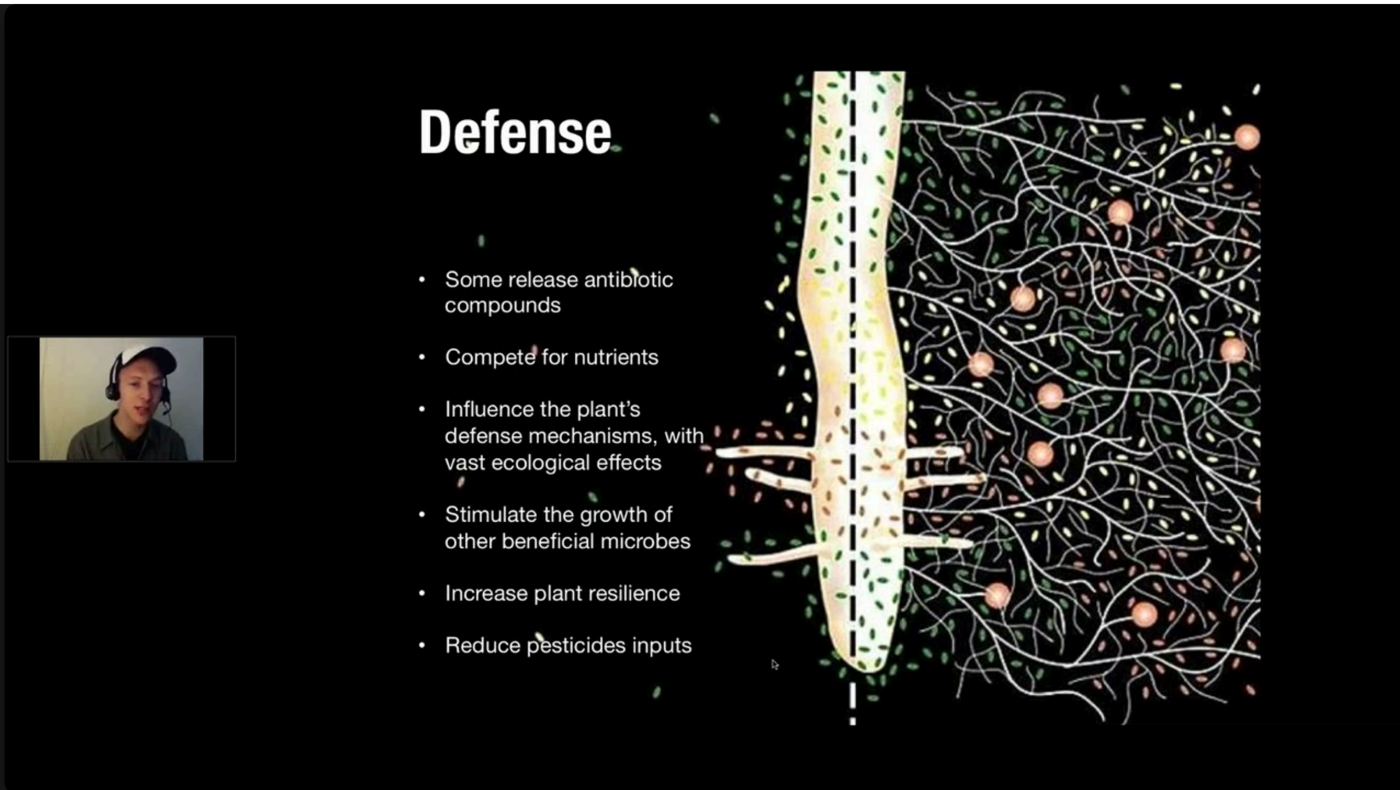
13. Defense



⏱ **31:06 – 32:51**

> Mycorrhizal fungi provide a lot of nutrients — lots of studies leading to higher nutritional content, medicinal quality, flowering rates, overall greater growth rates and survival time of the roots, and even shortening the time it takes the plant to reach maturity.
>
> They also do a lot to defend the plant — both as a physical barrier when the mycelium is enshrouding the root tip, but also by releasing all kinds of antibiotic compounds. They can release their own, they can influence the plant's own defense mechanisms — like with the insect example where the mycorrhizal fungi translate information to other plants in the system. The fungus itself might just compete for nutrients and out-compete competitors. It can also stimulate the growth of other beneficial microbes. Overall increasing plant health and reducing our need for pesticide inputs.

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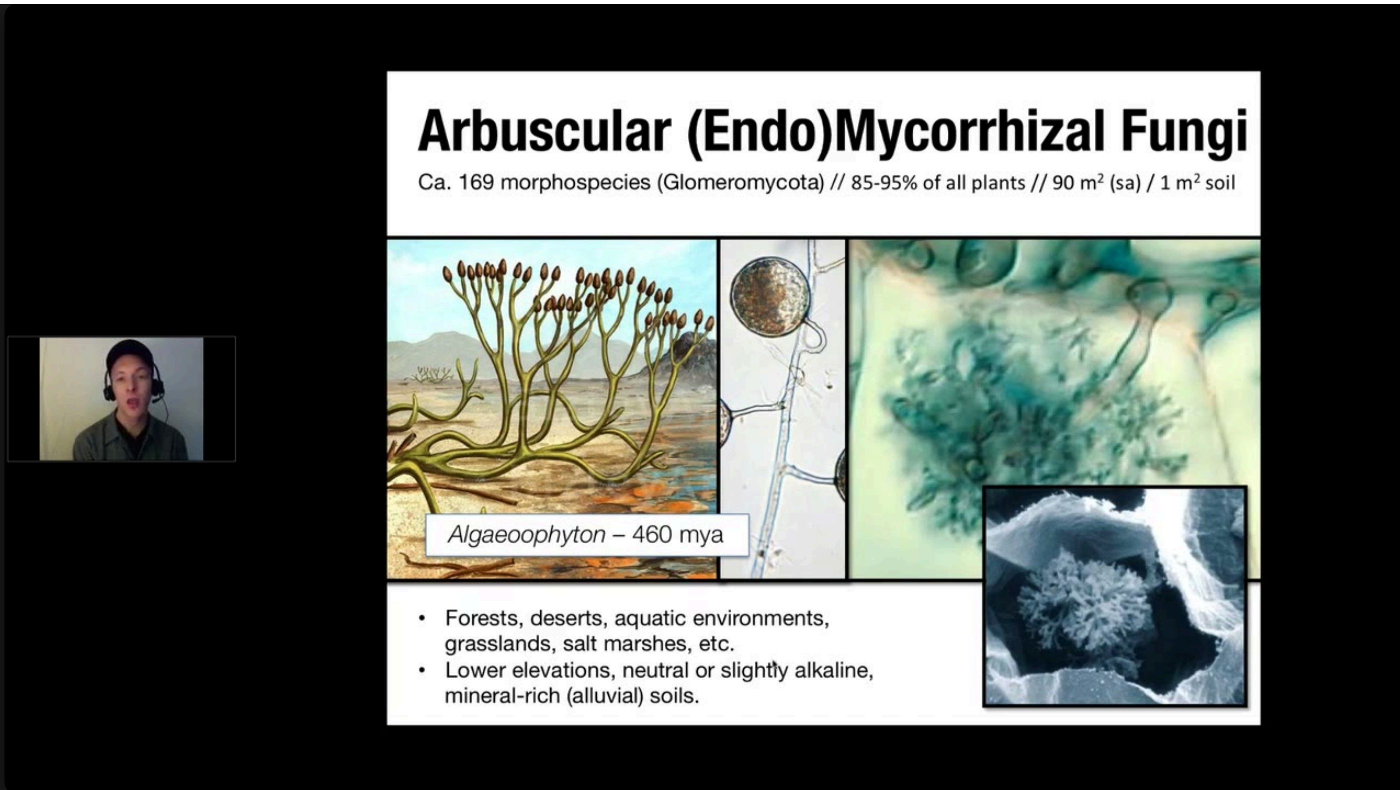
14. Arbuscular (Endo)Mycorrhizal Fungi



⏱ **32:51 – 36:06**

The arbuscular endomycorrhizal fungi or AM fungi — they are endos meaning they go into the cell wall. There are about 169 morphospecies in the group Glomeromycota. Many of them are ubiquitous around the world.

In the center we can see their large spores actually packed with hundreds or even tens of thousands of unique genetically distinct nuclei — which is fascinating and very strange biologically. But for our purposes this translates to the fact that they've got a lot of genetic potential and they can utilize that to adapt to their environment, basically shift whatever genetic expression they need based on what their environment demands. This kind of translates into not importing soil fungi — it's usually better practice to work with the local fungi because they're gonna be adapted to your environment.

On the left we're showing the first fossilized plant with roots from 460 million years ago — and here we find arbuscular mycorrhizal fungus on its roots that looks almost identical to species we find today. So it's a very ancient relationship, hasn't changed necessarily a ton over all these eons.

They're found on every continent except Antarctica — in forests, deserts, aquatic environments, grasslands, salt marshes — pretty much everywhere. Generally more alkaline soils, more mineral rich. They're the ones that really connect plants together — the generalists that don't discriminate — connecting 85 to 95% of all plants. They're super densely packed — in one square meter of soil their surface area might cover 90 square meters.

They've been shown to increase many crops' production rates. When the plants are stressed or starved they can provide up to 80 percent of the plant's phosphorus input. Their benefit can help us reduce our fertilizer input significantly. They can even reduce Fusarium and Phytophthora diseases. And they're a very important water source for desert plants.

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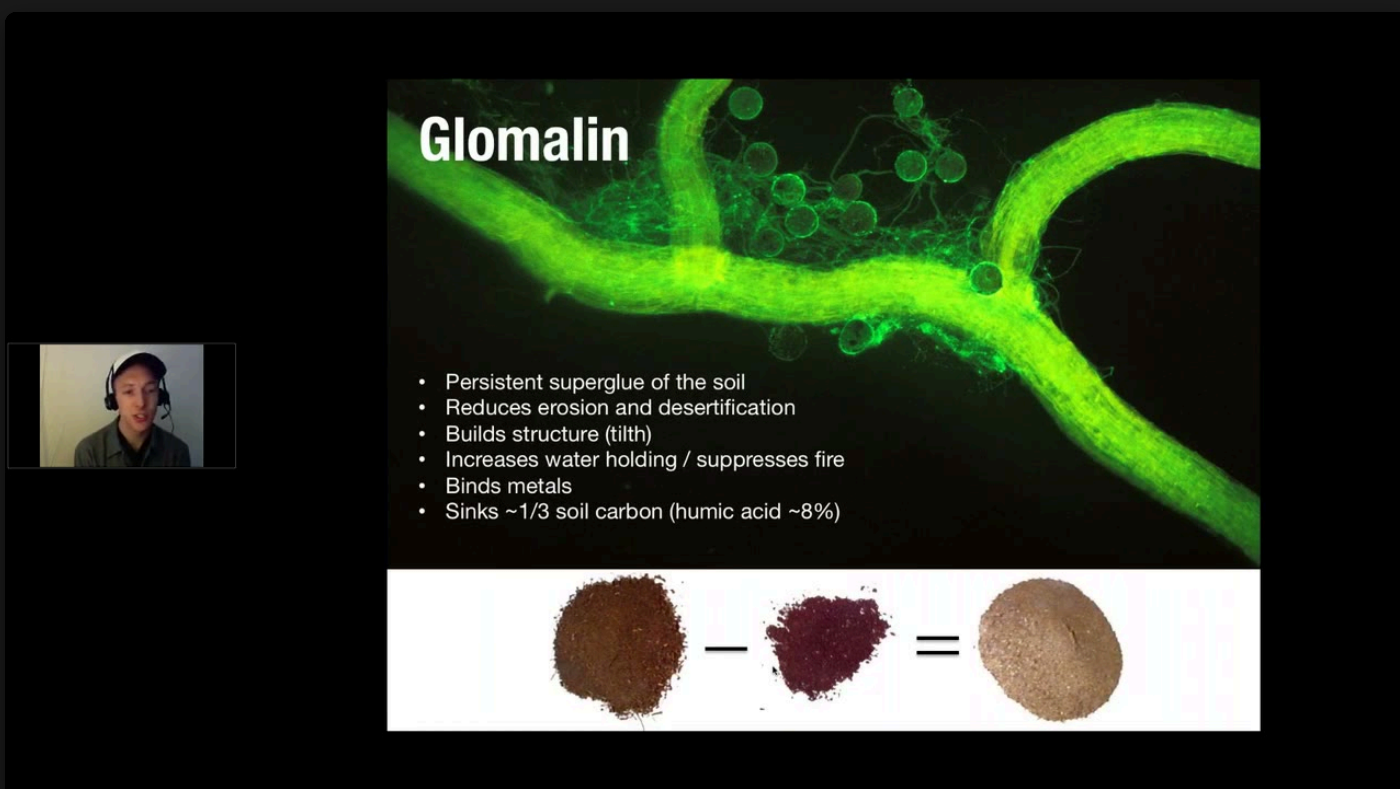
15. Glomalin



⏱ **36:06 – 37:54**

One of the most important things about the arbuscular mycorrhizae is they produce a unique sticky protein on the surface of their mycelium called glomalin. This is like the super glue of the soil — it's very persistent, lasting upwards of 50 years in the soil, likely due to its iron content. It is unique to these fungi and it's produced on their surface probably to hydrate and lubricate them.

But it also helps stick the soil together and basically creates macro aggregates that enable the greater tilth and structure in a soil that we all appreciate for water and air percolation, root movement, worm wiggling, etc. Really important for that aspect.

It also helps reduce erosion, desertification — thereby increasing water holding capacity and suppressing fires. Really integrally contributing to the structure and health of soil overall.

Remediation wise it can actually bind up some metals. And it's been argued — and actually shown — to be one of the major carbon sinks in the soil. It might actually be one of the biggest carbon sinks on its own right. When you take into account the mycelium itself which is a huge component of soil and huge carbon component — fungi and the glomalin might actually be one of the biggest if not the biggest carbon sinks in the soil. So we're talking about trying to sink carbon from the atmosphere — we need to think about getting more fungi going in our soils, not necessarily just trees.

Humic acids only account for about 8% of the soil carbon. On the bottom we're showing you can actually extract glomalin — it actually adds a lot of color and shading to the soil.

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16. AM Cultivation

⏱ 37:54 – 39:10

We can cultivate them. On the left I'm showing a photo from the Rodale Institute — they actually spent six, seven years working with farmers to figure out ways to grow a lot of AM cultivation inoculum cheaply and abundantly for farmers. You can look up the Rodale mycorrhizal cultivation protocol — it's pretty simple, straightforward. Basically grow grass or Allium or some quick growing plant in a pot or bag, and the roots and mycelium down below — harvest at the end of the season then apply to your crops the following year. Pretty straightforward, a lot of benefits — especially if we're working with the local fungi, local soils from a healthy forest, and amplifying that soil and those mycorrhizae.

On the upper right showing more technical processes that the West Virginia University is researching — they have a big arbuscular mycorrhizal research department. We can actually sieve your soil and separate out these spores, quantify their numbers and diversity — good for soil analysis, also good to track the efficacy of any attempts to cultivate these fungi, to increase their abundance in depleted soils. It's a nice metric to actually see how the fungal spore load is increasing and changing over time.

17. Monotropoid & Orchidaceous Mycorrhizae

⏱ 39:10 – 42:06

A couple other types to give a shout out to — the ericoid endomycorrhizal fungi are primarily restricted to Ericaceae species — blueberries and relatives, heather plants. They're more in the acidic soils these plants prefer — grasslands, tundras, heathlands, boreal forests. In some heathlands the soil can be pH 2.5, like lemon juice, and yet these fungi enable plants to survive. They can actually act as the primary decomposers of that whole ecosystem — the same plant they relate with dies and falls over, that fungus digests it and returns its nutrients to its brethren. These are called the drivers of these whole ecosystems.

All the monotrope plants that don't photosynthesize are fed by mycorrhizal fungi. They form another type of association — the fungus connects to these monotrope plants (the pink, the white, kind of strange looking plants in the forest) and then on the other end of that mycelial network connects to a tree and channels nutrients to the monotrope to feed it. An interesting fungus that does this is our king bolete or porcini.

On the right we have the orchid endomycorrhizae — all orchids form this when they germinate. All orchids do not photosynthesize initially — they need these fungi for their survival, sometimes for years, sometimes for the whole life of the orchid. Many common woodland mushrooms like Russula species do this. A really interesting one is the Armillaria or honey mushrooms — often seen as a pathogen, a parasitic mushroom — it actually feeds and supports a lot of orchids at the same time that it kills off some trees.

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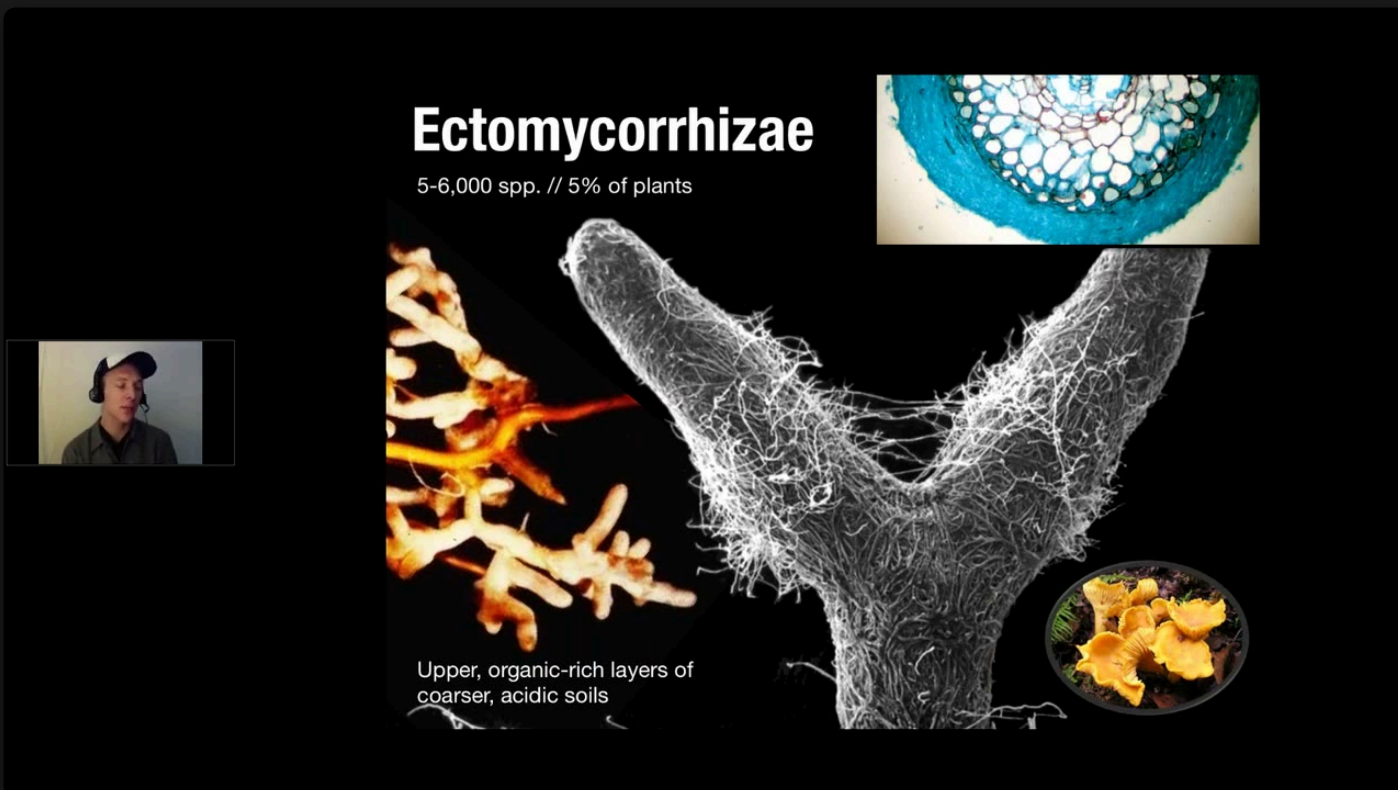
18. Ectomycorrhizae



⏱ **42:06 – 43:01**

> Our most common or probably most familiar mycorrhizal types of fungi are the ectomycorrhizae that form mushrooms and truffles. These form a thick layer of mycelium around the root tips — known as a mantle sheath. They're in the more organic-rich upper layers of soils, usually higher elevation, the older forests. This includes our chanterelles, boletes, gourmet truffles.
>
> These are the ones we wish we could cultivate — but as of yet only few species we've dialed in. Mostly ones that don't seem to be as dependent on really specific soil conditions. This is a big hurdle for us growers — trying to figure out the right soil conditions, microbial and insect

communities, other plant communities that say chanterelles need to fruit in our own backyard. Right now we can't do that consistently — that's why they are wild harvested.

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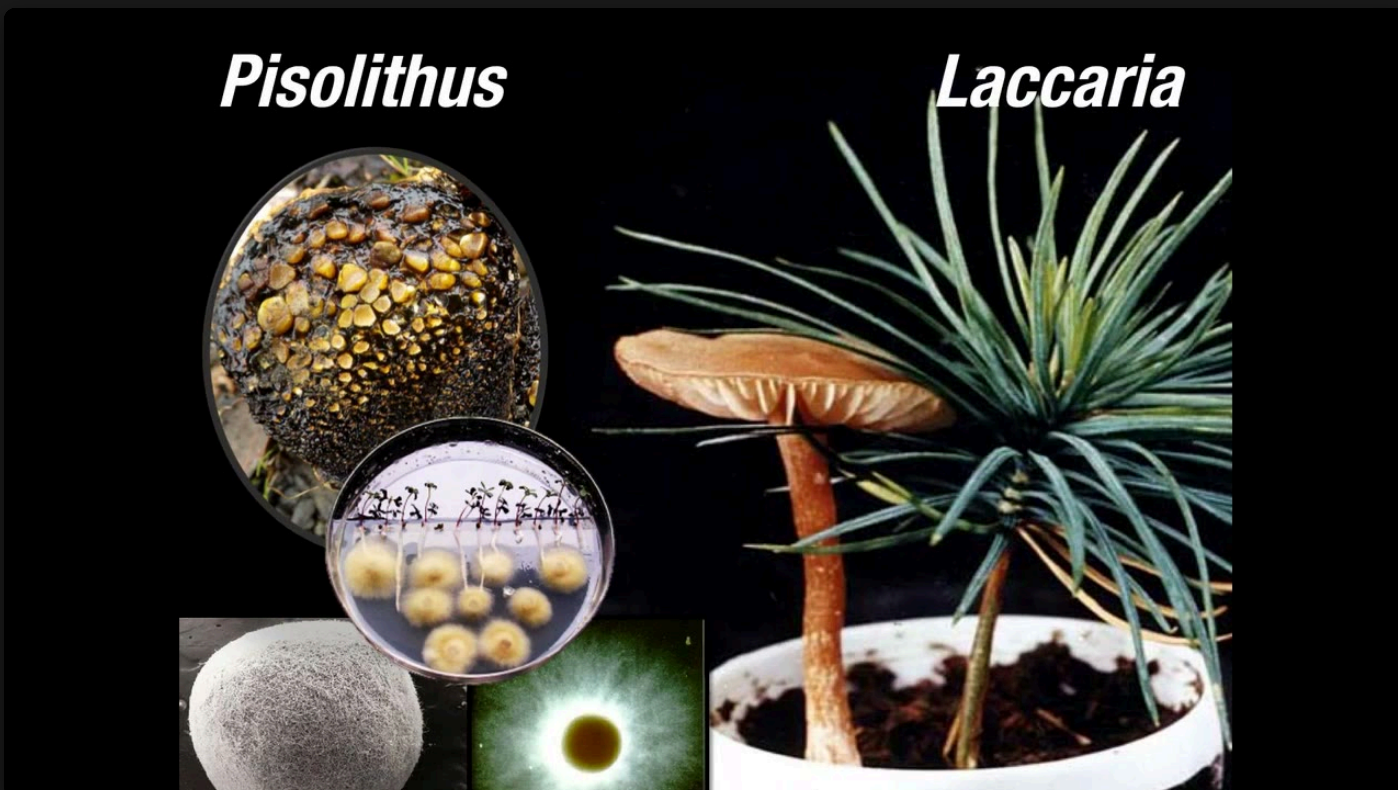
19. Pisolithus & Laccaria



⏱ 43:01 – 44:11

There are some species of ectomycorrhizae that we can cultivate — commonly cultivated for silviculture and tree plantations. Pisolithus tinctorius on the left — not so pretty but actually a very important mushroom to trees around the world, very heat tolerant, can survive in 104°F soils. Actually quite beneficial to trees in those kind of hotter environments. Establishes with many trees, very common species, establishes quite readily.

Other common species we can grow to inoculate trees are in the Rhizopogon, Hebeloma, and Scleroderma genera. Scleroderma is also important because it can associate with over 50 trees and tolerates soil pH anywhere from 2.6 to 8.4 — that's a huge range. An incredibly tolerant fungus, well adapted to helping trees in desert environments.

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20. Desert Truffles

⏱ **44:11 – 45:08**

Desert truffles can support a lot of plant species, especially Helianthemum shrubs. These desert truffles are a protein and medicine source and traditionally consumed by Bedouin cultures and many other cultures in North Africa, the Middle East, Mediterranean climates. Spoken about for a thousand years by the Prophet Mohammed.

We can cultivate them — it's not a common practice. I'm certainly interested in seeing more folks say in North America taking this on and seeing how viable it is as an industry and supplemental food crop — in say Southern California or Eastern Washington. I don't think anybody's really looking at it as far as I'm aware. Pretty interesting stuff possible there.

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21. Dark Septate (Class 4) Endophytes (DSEs)

⏱ 45:08 – 47:25

Dark septate endophytes — these are root-associated endophytic fungi. Endophytes live inside of plant tissue — all plants are permeated throughout their entirety by hundreds of trillions of fungi. These fungi are only found in roots. Interestingly they only associate or have a mechanistic state with the few plants that don't form true mycorrhizal associations — like our Brassica plants. They actually might be performing that mycorrhizal role while not being true mycorrhizae.

We can inoculate seeds with these types of fungi but studies are very limited at this point about what the benefit is. These are fungi that are found in those cryptobiotic crust communities in desert systems — they've actually been argued to help translate or translocate some of the fixed nitrogen that's created in these crusts to the other plants in the environment. This is known as the fungal loop hypothesis.

Also in soils we get other pathogens and fungi — I consider these the "vocal fungi." In many ways this is just a slight perception change — when we see these fungi doing quote-unquote bad things to our crops and woodlands, the question we need to ask is what did we do to bring this about as humans. Most of the great fungal blights of history were caused by human mismanagement of an ecosystem — monoculturing being an extreme example where we strip away endophytic and mycorrhizal diversity, the plants are left defenseless and sure enough something comes in to reset the scales.

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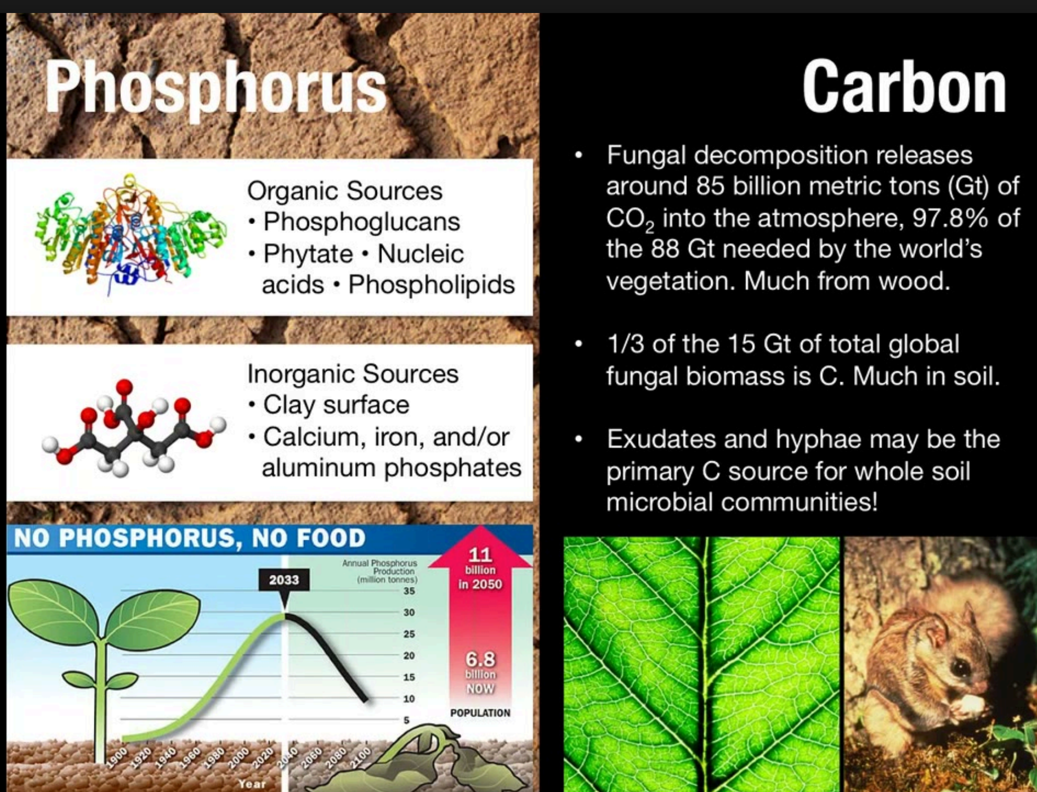
22. Phosphorus & Carbon Cycles

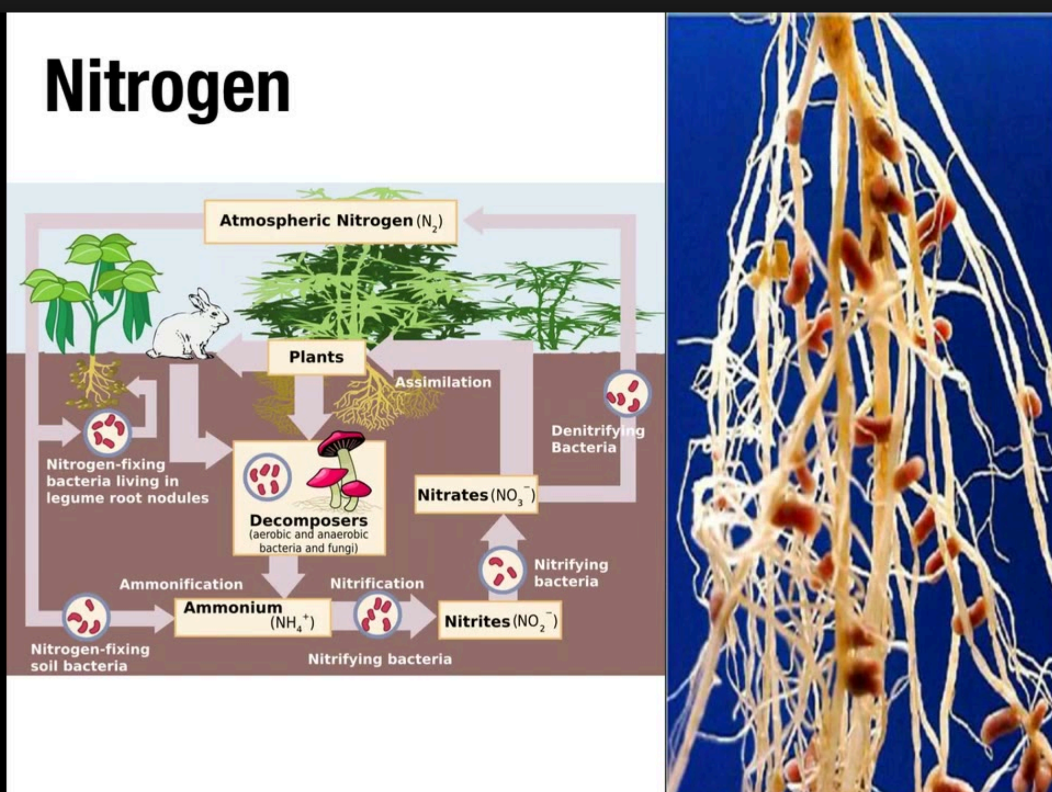


⏱ **47:25 – 48:57**

Fungi are integral to nutrient cycling. For the phosphorus cycle — they can break down organic matter, decompose it through their phosphatases from organic sources. With their strong acids they can release inorganic phosphorus from clays and also mineral phosphates. We are of course facing peak phosphorus in the future — so reducing our extraction of phosphorus from unsustainable sources is going to be important. Bringing these mycorrhizal fungi and other microbes to help naturally do this is going to be really important now and certainly into the future.

For the carbon cycle — fungi decomposing provide almost 98% of the carbon dioxide needed for the vegetation in the world. Humans and other animals provide less than about 2% of the rest. About a third of all fungal biomass is carbon and much of that is stored in soils. Again with the glomalin that's a major soil carbon sink. And that might be one of the primary — or might even be the primary — carbon source for whole soil microbial communities.

23. Nitrogen Cycle



⏱ 48:57 – 51:04

For the nitrogen cycle — fungi help release it through normal decomposition of organic matter. Many ectomycorrhizal mushrooms will scavenge for ammonium — they don't usually prefer nitrate — so they'll actually help sort of gather that for the plants which also prefer ammonium generally speaking. They tend to dominate in nitrogen-poor soils whereas the arbuscular mycorrhizae tend to predominate in phosphorus-poor soils.

Some species like Laccaria mushrooms actually can eat insects — and in one study they found up to 25 percent of the nitrogen in a Laccaria-associated tree was actually provided from these insects. So even just the consumption of insects can be a major nitrogen source for plants.

Arbuscular mycorrhizae actually significantly contribute to the nodulation and association of rhizobial nitrogen-fixing bacteria on leguminous plants. They work synergistically in a tripartite relationship. The plant, the fungus provides phosphorus, carbon and zinc that it brings in through its network which assists in the nodulation and nitrogen fixation process. In exchange the bacteria enhances the phosphorus uptake of the fungus and the plant with its own organic acids and phosphatases. The bacteria seem to influence the spore germination, even the growth rate of the fungus. At the same time the fungus releases what's known as the Myc factor which helps activate the rhizobia formation of the nodulation — and together they form a beneficial biofilm in the mycorrhizosphere.

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24. Lichens

⏱ 51:04 – 52:29

> Lichens are actually an important part of soils. Forests start out from bare rock and usually the first thing showing up on that bare rock are lichens — just like in the early Earth. The fungal acids produced by the 95% fungal component of a lichen release those minerals, enable mosses to grow, eventually leading to the climax forest.
>
> In that forest the lichens can actually provide up to 50% of the nitrogen input just through their own nitrogen fixation. The darker types of lichens are primarily photosynthesizing due to cyanobacteria — cyanobacteria can fix nitrogen. So when that lichen falls and decomposes it releases that to the soil along with the minerals that it collects out of the atmosphere — probably a big mineral input into soils due to lichens trapping that out of the atmosphere and falling down and releasing that when they decompose.

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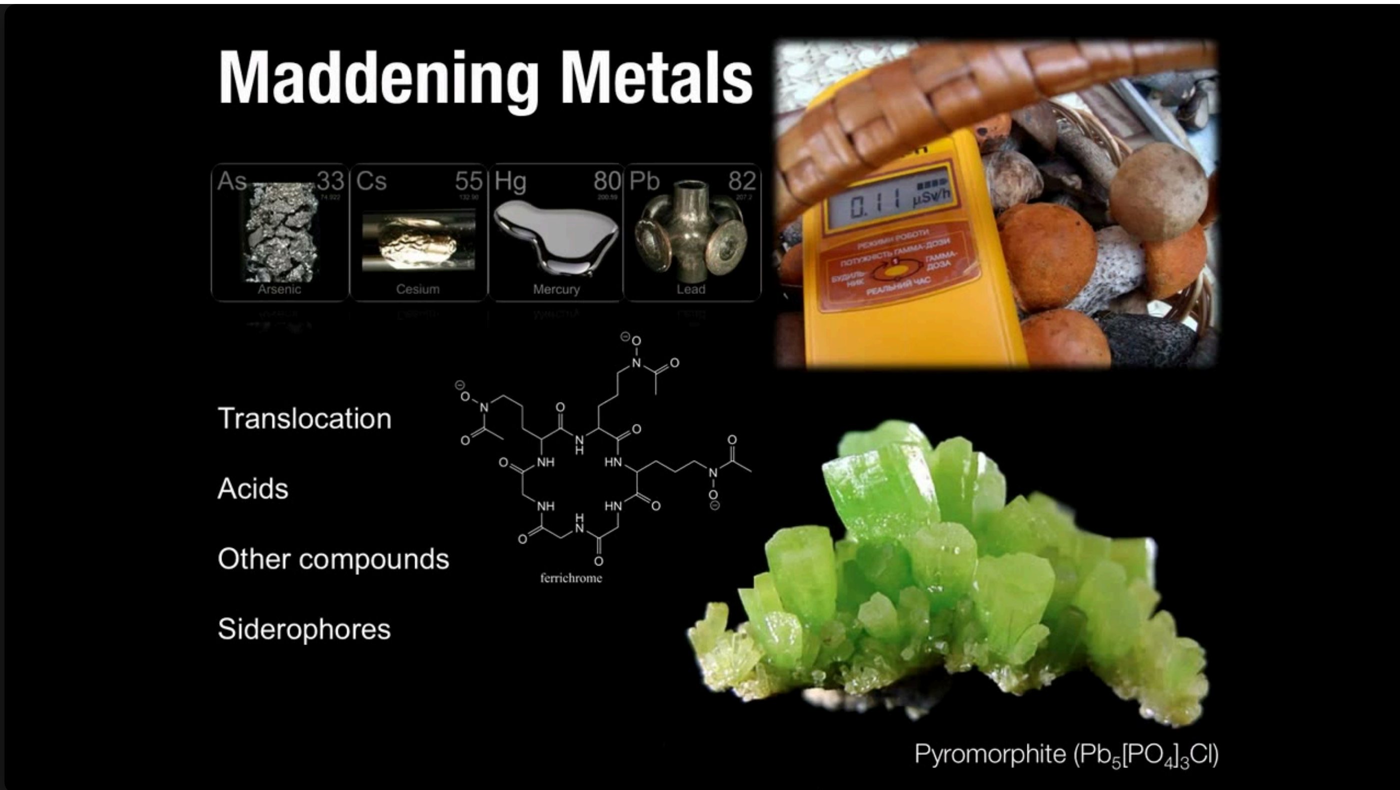
25. Maddening Metals



⏱ 52:29 – 54:50

Lastly we'll talk about the remediative roles of fungi. There's two main ways that fungi can help deal with polluted soils. On one hand is with the heavy metals — which of course are elements on the periodic table that can't be turned into something else. All fungi can do is either move them around or sort of lock them up or bind them up so that they don't harm or interfere with other organisms.

Just like plants, many mushrooms can accumulate these metals out of the soil and concentrate them into their fruit body. The question then becomes what do we do with the heavy metal mushroom — just like the question with sunflowers. Unfortunately we don't have a good answer to that on the backyard scale. Industrially they would incinerate the plant or mushroom in a way to trap the metals in the ashes. Generally we have to just dispose of it as toxic material — but at least we are scrubbing the soil slowly. This can take generations, not a one-season process.

In the upper right we're seeing Leccinum mushrooms still accumulating radioactive cesium outside Chernobyl almost 40 years later — so it's a very slow generational process but fungi do it.

Fungi can also release acids and other compounds — even glomalin — which combine with metals and sort of stabilize them for an unknown amount of time. They sort of just stay there in complexes. Lastly some fungal components on their tissue can actually bind to metals naturally — when we put mycelium into say water systems contaminated with metals, the metals might bind to the mycelium and sort of pull it out.

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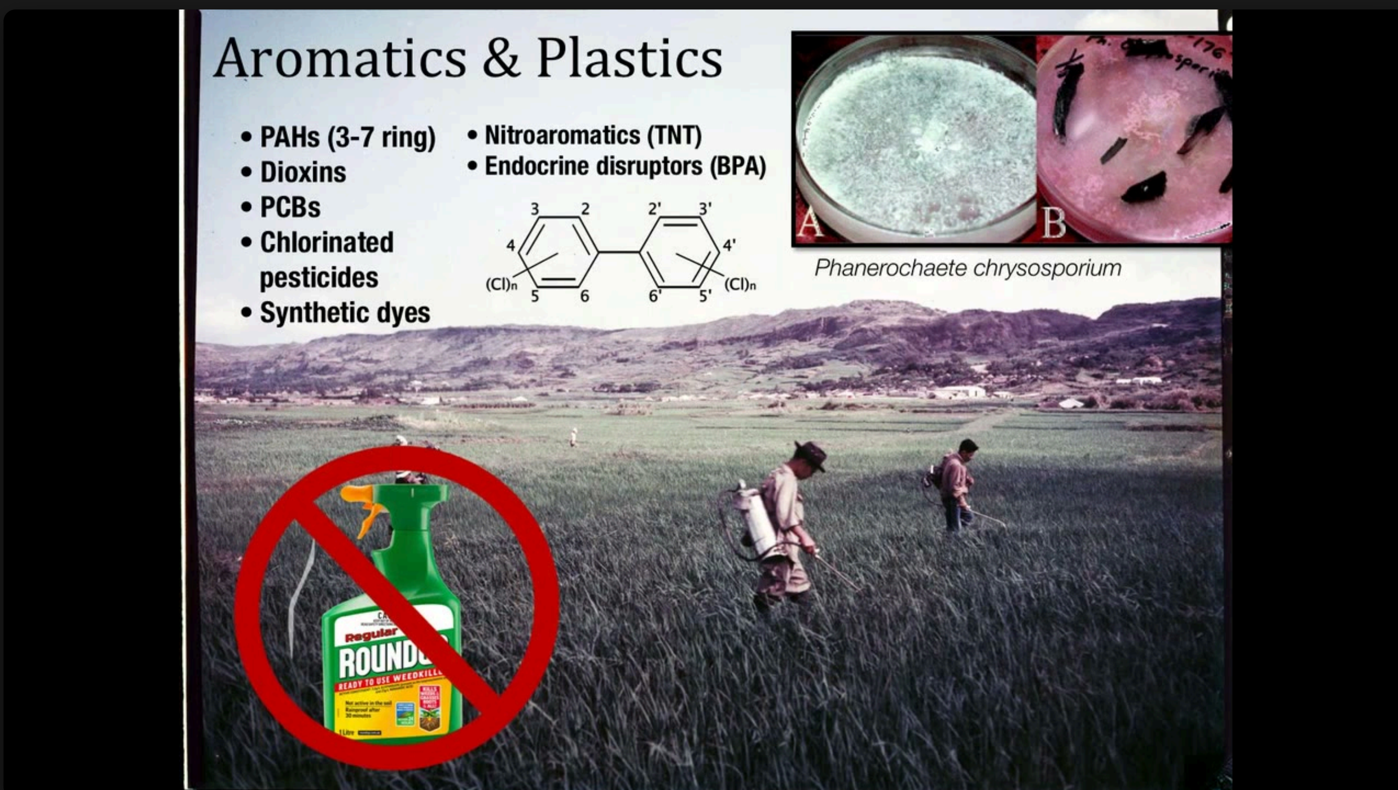
26. Aromatics & Plastics



⏱ 54:50 – 56:43

Fungi can break down all kinds of chemicals and this has been well researched for some 50 years. Initially starting out with wood-decomposing fungi — these fungi are able to break down the most complex thing in nature, so some smart person said hey maybe they can break down all the complex chemicals we're inventing as humans. Turns out wood decomposers can break down many of these chemicals — but so can many molds.

The Trichoderma molds I mentioned can break down glyphosate — meaning Roundup — DDT, many other herbicides naturally. So even simple molds can do this work. Many ectomycorrhizal mushrooms and even arbuscular mycorrhizal fungi can break down many chemicals — so it seems to be a natural trait of many fungi. Even fungi harvested from pristine environments can break down herbicides naturally.

The challenge now that we face is how do we translate what we've learned in the lab — where most of this work has been done — into field, vast applications and protocols that are consistent, safe, and measurable.

Different types of fungi can break down even plastics. For the hundred years of plastics' existence, many studies have been done where they buried plastics in the soil — and time and time again every single time they dig up the plastic years later it's being degraded, slowly broken down, especially by fungi.

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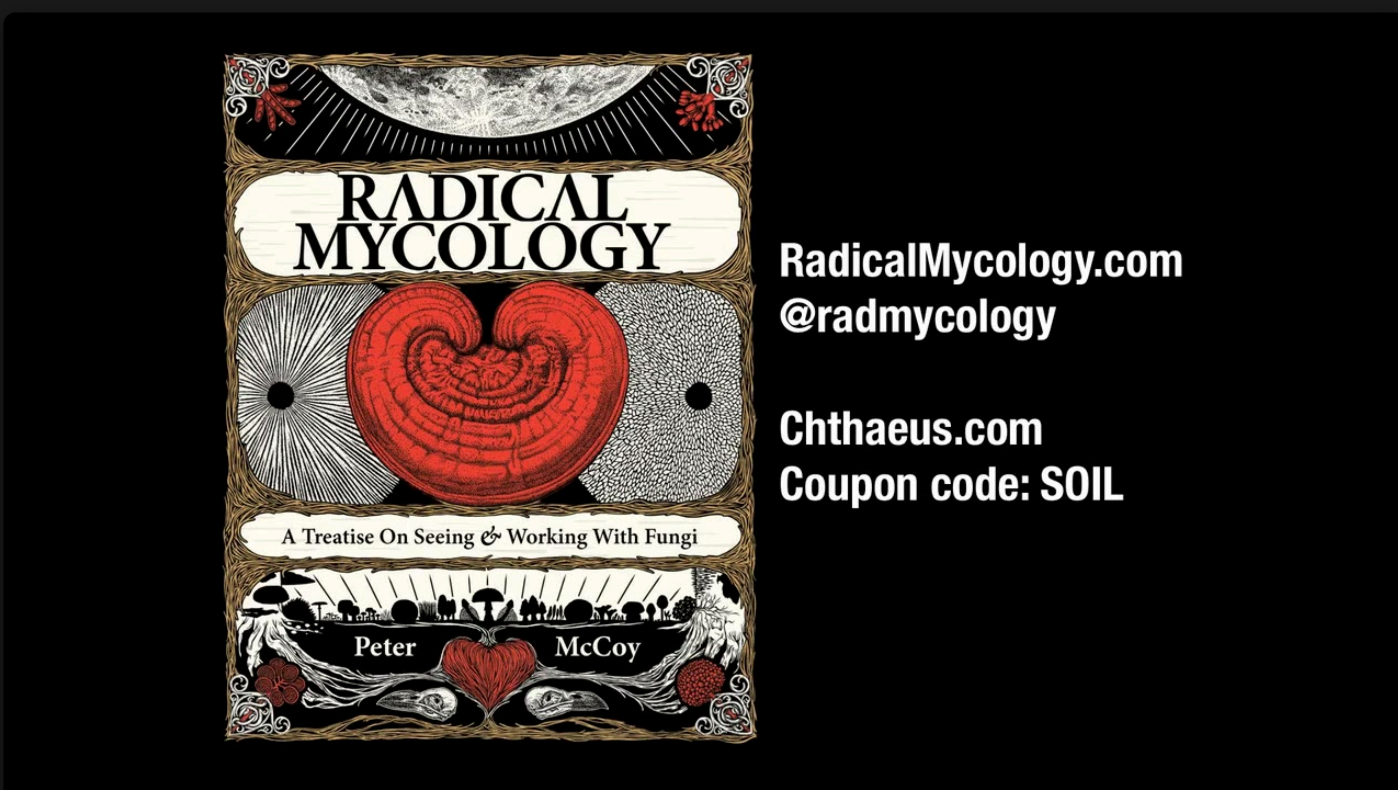
27. Radical Mycology – Book & Resources



⏱ 56:43 – 59:30

Fungi can be incorporated in a myriad of ways into our designs — as permaculturists or just good farming practices or even just trying to have a garden in your backyard. So often when I read books on these types of topics the first thing I'll do is flip to the index and see how many entries are there for fungi, mycelium, mushrooms — and more often than not there's none or maybe just a couple and they're very cursory.

I point that out just to challenge you to always ask yourself — how can fungi fit into this design? Even if the book or whoever you're talking to hasn't said it explicitly. Because it's not necessarily on their radar — now it's on yours and you can start to think hmm, what are fungi doing in this system, what's missing from this analysis, how could fungi be incorporated?

Ultimately cultivating fungi and working with them isn't too complex. Biology is unfathomably complex but at the same time it's relatively simple — a seed put in the soil sprouts, fungi are kind of straightforward in the same way when you understand them.

So hopefully this talk gave you a good whirlwind introduction to the soil fungal communities and all the things they do. As a thank you for watching this talk — I'm offering a discount on my book at Chthaeus.com. Enter the coupon code SOIL when you checkout — you get five bucks off the book for the next 24 hours. Otherwise you can check out RadicalMycology.com for some free videos and resources. We're on social media — @radmycology is the handle.

Riley: That was fantastic man — totally a firehose of amazing info. The stuff on glomalin — that blew my mind. I had no idea that was the biggest source of carbon.